

Programme	B. Sc. STATISTICS				
Course Code	STA 8 EJ 413				
Course Title	QUEUEING MODELS				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Basic knowledge of Markov Chain & Stochastic process				
Course Summary	Detail analysis of Queueing Models				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
1.	Identify basic concepts of queueing theory	U	F	Seminar Presentation
2.	Analyze behaviors of queueing models	An	P	Instructor-created exams
3.	Explain on queueing networks	Ap	F	Group Tutorial Work
4.	Apply queueing models	Ap	F	Home Assignments
5.	Evaluate performance measures	An	P	Practical Skill
6.	Create significance and applications of queueing theory	C	M	Group Tutorial Work
* - Remember (R), Understand (U), Apply (Ap), Analyze (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (60)	Marks (70)
I	Queueing Theory		12	20
	1	Introduction to queueing theory, Cost Equations, Steady-State Probabilities		
	2	Characteristics of queueing processes, Measures of effectiveness		
	3	Markovian queueing models		
	4	steady state solutions of the M/M/I model, waiting time distributions		
	5	Little's formula, queues with unlimited service, finite source queues		
II	Transient Behavior		12	15

	6	Transient behavior of M/M/1 queues		
	7	Transient behavior of M/M/ ∞		
	8	Busy period analysis for M/M/1 and M/M/c models		
	9	Advanced Markovian models		
	10	Bulk input M ^[X] /M/1 model, Bulk service M/M ^[Y] /1 model		
	11	Erlangian models, M/Ek/1 and Ek/M/1		
	12	A brief discussion of priority queues		
III	Queueing Networks		12	20
	13	Queueing networks-series queues		
	14	Open Jackson networks		
	15	Closed Jackson network		
	16	Cyclic queues		
	17	Extension of Jackson networks		
	18	Non Jackson networks		
IV	General Queueing Models		12	15
	19	Models with general arrival pattern, The M/G/1 queueing model		
	20	The Pollaczek-khintchine formula, Departure point steady state systems size probabilities, ergodic theory		
	21	Special cases M/Ek/1 and M/D/1, waiting times, busy period analysis, general input and exponential service models,		
	22	Arrival point steady state system size probabilities		
V	Open –Ended		12	
	Problems regarding Module I to Module IV			

Reference

1. **Gross, D. and Harris, C.M.(1985).** Fundamentals of Queuing Theory, 2nd Edition, John Wiley and Sons, new York.
2. **Kleinrock L (1975).** Queuing Systems, Vol. I &Vol 2, John Wiley and Sons, New York.
3. **Ross, S.M. (2007).** Introduction to Probability Models. 9th Edition, Academic Press, New York.
4. **Bose, S.K. (2002).** An Introduction to Queuing Systems, Kluwer Academic/Plenum Publishers, New York.